Chapter 12. Generalized Least Squares

Correlated and heteroscedastic errors, GLS estimator, properties

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12.1 The general linear model

So far the errors were uncorrelated with equal variance. In time series, clustered or heteroscedastic data this fails. The **general linear regression model** is

$$\mathbf{y} = X\beta + \epsilon, \qquad \epsilon \sim N_n(\mathbf{0}, \sigma^2 V),$$

with V a known positive definite matrix, X of size $n \times (p+1)$ and full column rank. Ordinary least squares (OLS) is no longer the BLUE.

Since V is symmetric positive definite, $V = P\Lambda P'$ with P orthogonal, and $T = P\Lambda^{1/2}$ satisfies $V = TT'$. Multiply the model by T^{-1}:

$$\mathbf{z} = W\beta + \eta, \qquad \mathbf{z} = T^{-1}\mathbf{y}, \quad W = T^{-1}X, \quad \eta = T^{-1}\epsilon,$$

and $\operatorname{Var}\eta = \sigma^2 T^{-1}V(T^{-1})' = \sigma^2 T^{-1}TT'(T')^{-1} = \sigma^2 I_n$. The transformed model satisfies the usual assumptions.

Theorem 12.1 (Generalized least squares)

The BLUE of β is the **generalized least squares** (GLS) estimator

$$\hat\beta_G = (W'W)^{-1}W'\mathbf{z} = (X'V^{-1}X)^{-1}X'V^{-1}\mathbf{y},$$

which minimizes $(\mathbf{y}-X\beta)'V^{-1}(\mathbf{y}-X\beta)$. It satisfies $E\hat\beta_G = \beta$ and $\operatorname{Var}\hat\beta_G = \sigma^2(X'V^{-1}X)^{-1}$.

Proof. Apply the Gauss–Markov theorem (Theorem 7.1) to the transformed model; linear unbiased estimators in $\mathbf{z}$ are the same as those in $\mathbf{y}$, since T is invertible. Using $(T^{-1})'T^{-1} = (TT')^{-1} = V^{-1}$, $W'W = X'V^{-1}X$ and $W'\mathbf{z} = X'V^{-1}\mathbf{y}$. Also $\|\mathbf{z}-W\beta\|^2 = (\mathbf{y}-X\beta)'V^{-1}(\mathbf{y}-X\beta)$. Finally $E\hat\beta_G = (X'V^{-1}X)^{-1}X'V^{-1}X\beta = \beta$ and $\operatorname{Var}\hat\beta_G = (X'V^{-1}X)^{-1}X'V^{-1}(\sigma^2V)V^{-1}X(X'V^{-1}X)^{-1} = \sigma^2(X'V^{-1}X)^{-1}$. ∎

When V is diagonal, $V = \operatorname{diag}(v_i)$, GLS is **weighted least squares** with weights $w_i = 1/v_i$: minimize $\sum w_i(y_i - \mathbf{x}_i'\beta)^2$, giving less weight to noisier observations.

12.2 Properties

OLS under correlated errors. The OLS estimator $\hat\beta = (X'X)^{-1}X'\mathbf{y}$ is still unbiased, but

$$\operatorname{Var}\hat\beta = \sigma^2(X'X)^{-1}X'VX(X'X)^{-1},$$

which is at least $\sigma^2(X'V^{-1}X)^{-1}$ (by Gauss–Markov), with equality when $V = I$ (more generally, when the column space of X is invariant under V).

Residuals and σ^2. The GLS residual vector is $\mathbf{e}_G = \mathbf{y} - X\hat\beta_G = \{I - X(X'V^{-1}X)^{-1}X'V^{-1}\}\mathbf{y}$, with

$$E\mathbf{e}_G = \mathbf{0}, \qquad \operatorname{Var}\mathbf{e}_G = \sigma^2\{V - X(X'V^{-1}X)^{-1}X'\}.$$

Theorem 12.2

$SSE(\hat\beta_G) = \mathbf{e}_G'V^{-1}\mathbf{e}_G$ satisfies $E\,SSE(\hat\beta_G) = (n-p-1)\sigma^2$, so

$$\tilde\sigma^2 = \frac{(\mathbf{y}-X\hat\beta_G)'V^{-1}(\mathbf{y}-X\hat\beta_G)}{n-p-1}$$

is unbiased for σ^2, and $\widehat{\mathrm{Var}}(\hat\beta_G) = \tilde\sigma^2 (X'V^{-1}X)^{-1}$.

Proof. $SSE(\hat\beta_G)$ is the residual sum of squares of the transformed model, $\|\mathbf{z} - W\hat\beta_G\|^2$, whose expectation is $(n-p-1)\sigma^2$ (Theorem 6.2). Directly: by Theorem 5.4(a),

$$\begin{aligned} E(\mathbf{e}_G' V^{-1}\mathbf{e}_G) &= \sigma^2 \mathrm{tr}[V^{-1}\{V - X(X'V^{-1}X)^{-1}X'\}] \\ &= \sigma^2\{n - \mathrm{tr}((X'V^{-1}X)^{-1}X'V^{-1}X)\} = \sigma^2(n-p-1). \end{aligned}$$

■

By contrast, the OLS estimate $\hat\sigma^2 = \mathbf{y}'(I-H)\mathbf{y}/(n-p-1)$ has

$$E\hat\sigma^2 = \frac{\sigma^2}{n-p-1}[\mathrm{tr}V - \mathrm{tr}\{(X'X)^{-1}X'VX\}],$$

which in general differs from σ^2, so the usual OLS standard errors $\hat\sigma^2(X'X)^{-1}$ are biased in two ways under correlated errors.

Under normality, all the inference of Chapters 6–8 carries over to the transformed model: for instance $\hat\beta_G \sim N(\beta, \sigma^2(X'V^{-1}X)^{-1})$, $(n-p-1)\tilde\sigma^2/\sigma^2 \sim \chi^2(n-p-1)$, and t and F tests use $\tilde\sigma^2$.

Example 12.1 (Averages with unequal precision)

Suppose $y_i = \mu + \epsilon_i$ where y_i is the average of m_i measurements, so $\mathrm{Var}\,\epsilon_i = \sigma^2/m_i$ and $V = \mathrm{diag}(1/m_i)$. Then $X = \mathbf{1}$ and

$$\hat\mu_G = \frac{\sum m_i y_i}{\sum m_i}, \qquad \mathrm{Var}\,\hat\mu_G = \frac{\sigma^2}{\sum m_i},$$

the grand average of all measurements. The OLS estimate $\bar y$ has variance $\frac{\sigma^2}{n^2}\sum \frac{1}{m_i} \geq \frac{\sigma^2}{\sum m_i}$ by the harmonic–arithmetic mean inequality.

Example 12.2 (AR(1) errors)

If $\epsilon_t = \rho\epsilon_{t-1} + u_t$ with $|\rho| < 1$ and u_t i.i.d. $N(0,\sigma_u^2)$, then $\mathrm{Var}\,\epsilon = \frac{\sigma_u^2}{1-\rho^2}V$ with $V_{st} = \rho^{|s-t|}$. The transformation T^{-1} amounts to the familiar quasi-differencing $y_t - \rho y_{t-1}$, $\mathbf{x}_t - \rho\mathbf{x}_{t-1}$ for $t \geq 2$, with the first observation scaled by $\sqrt{1-\rho^2}$. In practice ρ is unknown and is estimated from OLS residuals (feasible GLS).

Unknown V. V is rarely known. It is modelled with few parameters (equal correlation within clusters, AR(1) in time, variance proportional to a power of a regressor) and estimated from the data, typically from OLS residuals; GLS with the estimate plugged in is **feasible GLS**.